Predicting New York Taxi Demand and Fares - Group 22

This report will explore the factors influencing Manhattan taxi demand and average fare amounts and explain how mathematical models can be used to predict these quantities a day in advance of operation. We briefly outline the data processing that was used in order to prepare the dataset, as well as the modelling process and the finalised models. We also outline time-based and geographical patterns found during the development process; and offer recommendations to improve driver allocation, operational efficiency, and profitability.

1 The Dataset

The dataset is obtained from the *NYC Taxi and Limousine Commission* and consists of yellow taxi trip records from 2024. After filtering to Manhattan pickup locations, the dataset contains over 27.5 million trips. Given its size, initial analysis and model development were performed on a representative subset before scaling to the full dataset. As demand patterns vary throughout the year, a random week from each month is excluded from analysis and used to objectively evaluate the model performances. All the remaining data (approximately 75%) is used for training and analysis.

2 Data Processing and Exploration

2.1 Features and Data Cleaning

The dataset initially contains 19 fields of information (features) per trip, including time-based, geographical, and fare-related information.

Initial exploration reveals the presence of implausible observations, such as negative costs, extremely short durations leading to unrealistic speeds, and disproportionate trip distances. Rather than attempting to correct all such cases, the approach focuses on removing extreme outliers that could negatively impact model performance.

To achieve this, the following filtering rules are applied:

- Passenger count, extra charges, and airport fee must be non-negative.
- Trip distance must be greater than 0 and no more than 40 miles, and average speed must be greater than 0 and no more than 40 mph.
- Fare amount must be between 0 and 100, fare per hour between 0 and 1000, and total amount between 4.00 and 250.
- Tip amount and tolls amount must be between 0 and 40.
- MTA tax must be between 0 and 1, and improvement surcharge must equal 1.0.
- Congestion surcharge must be either 0 or 2.5 (missing values are allowed).
- Trip duration must be between 0.01 and 2 hours and average speed between 0 and 40 mph.

These choices for the cut-off points were justified by analysis of NYC yellow taxi policies, available on the TLC website. Cleaning removes less than 4% of the data. This is relatively high, indicating a noisy dataset, but given the size of the dataset, this is acceptable. This means that modelling can proceed on a dataset that is generally representative of realistic operation.

2.2 Missing Data

The dataset is largely complete, although approximately 10% of observations are missing values in fields such as passenger count, RatecodeID, congestion surcharge, and airport fee. This is primarily driven by differences in reporting across vendors. In particular, **Creative Mobile Technologies, LLC** and **Curb Mobility, LLC** exhibit partial missingness (around 10%) in these fields, while **Myle Technologies Inc** does not record them at all. As a result, the missing data is systematic rather than random. If the models were to use any of these features, the missing values would have to be substituted for sensible values. However, ultimately none of these features were progressed to the modelling stage, so this was not necessary.

3 Aggregating the Dataset

To understand demand and average fares in a practical way, trips are aggregated (grouped) by taxi zone and by hour of pickup. Using hourly intervals strikes a balance between capturing meaningful changes throughout the day and ensuring there is enough data in each group to give reliable results. Pickup time is used (rather than drop-off time) because it reflects when demand actually occurs — that is, when drivers are needed.

3.1 Target Variables

Two target variables are used: **total trip count per zone-hour** (demand) and **average fare per trip per zone-hour**.

Alternative measures (such as fare per mile or per hour) were considered, but average fare per trip is easier to interpret, as it reflects the additional income generated by each extra journey. This makes it a useful way to compare earning potential across different locations and times.

Total revenue per zone-hour was not used as a target, as it is largely driven by demand and therefore provides limited additional insight. Additionally, we used "fare amount" instead of "total amount" as the latter includes fixed fees and tips making predictions more difficult.

3.2 Model Features

After aggregating the data, we create a large set of features that describe both recent activity and repeating patterns over time. We allow the model to "look back" at what has happened previously in order to help predict the future - such features are called *lag* features. Some of the features we included in the models are:

- Lagged target variables (capturing recent and daily temporal patterns)
- Lagged aggregate features (e.g. distance, duration, speed)
- Categorical time features (hour of day, day of week)
- Location-based features (zone identifiers)
- Trip composition features (e.g. ratecode proportions and their lags)

These features allow the models to capture both short-term trends (e.g. daily patterns) and longer-term periodic behaviour (e.g. weekly cycles). Importantly, in order to capture changing rush hour times, a 23h lag feature is also included. In practice, this means the models can be run for a given hour as soon as data for that hour is collected the day before, allowing time to adapt driver allocation and instructions to the changing conditions.

4 Model Selection

To ensure robust performance, three models are developed for both demand and revenue prediction: a linear regression model (baseline), a gradient boosting model, and a neural network. This combination allows for comparison between simple, interpretable models and more flexible, but expensive, approaches.

5 Model Results and Comparison

5.1 Initial Demand Model Results

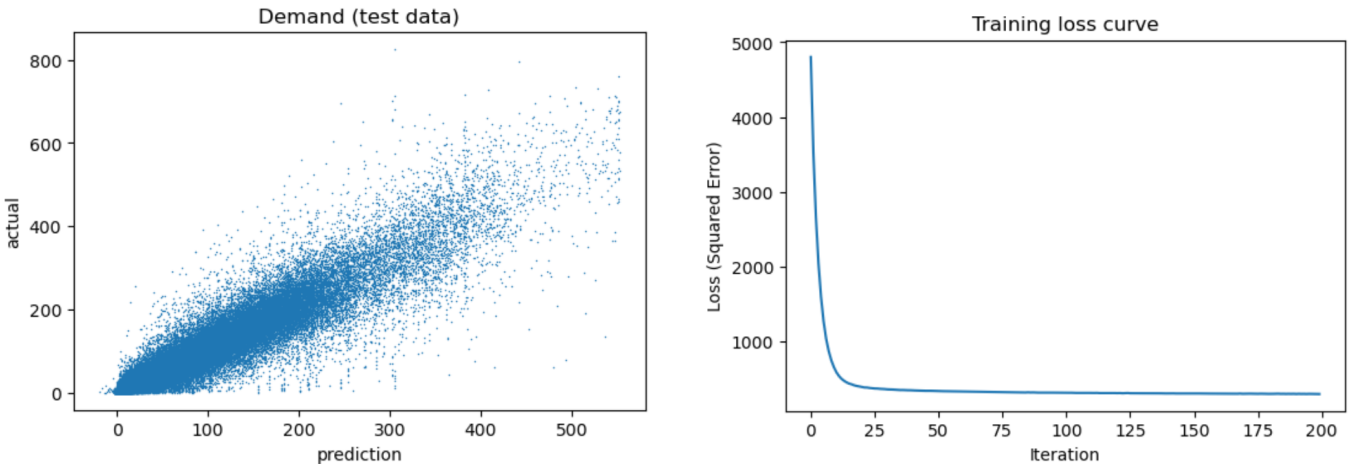
Model	Typical Error (Trips per Zone-Hour)
Linear Regression	46 ± 5
Gradient Boosting	31 ± 5
Neural Network	26 ± 6

Table 1: Average prediction error for trip count per zone-hour (Mean RMSE and STD from cross validation)

As shown in Table 1, the Neural Network provides the most accurate predictions, with a typical error of around 26 trips per zone-hour. This is a clear improvement over Gradient Boosting (around 31 trips) and Linear Regression (around 46 trips), meaning its predictions are consistently closer to the true demand. While the Neural Network's

performance varies slightly more across different samples, the gain in accuracy is substantial. For this reason, it is selected as the final model.

5.2 Final Demand Model



(a) Demand Model Actual vs Predicted Plot

(b) Training Loss Plot

Figure 1

Dataset	Typical Error (Trips)	Avg. Error (Trips)
Train	24	14
Test	31	17

Table 2: Prediction accuracy of the final model (trips per zone-hour). Typical error refers to RMSE and average error refers to MAE

The model performs strongly overall and maintains similar accuracy on new data (Table 2), indicating it can be relied on in practice. On unseen data, predictions are within around 17 trips per zone-hour of actual demand (on average). This level of accuracy is sufficient to support operational decisions such as driver allocation. Errors increase slightly when moving from training to test data (from around 24 to 31 trips typical error), which is expected and suggests that the model generalises to unseen data reasonably well. As shown in Figure 1a, the model performs best in high-demand areas, where predictions are more stable. Accuracy is lower in low-demand zones (fewer than 10 trips per hour), where demand is naturally more variable. Overall, the model captures the key demand patterns effectively and generalises well to new data, although results will be more reliable for busy zone-hours than those without many trips.

5.3 Initial Average Fare Model Results

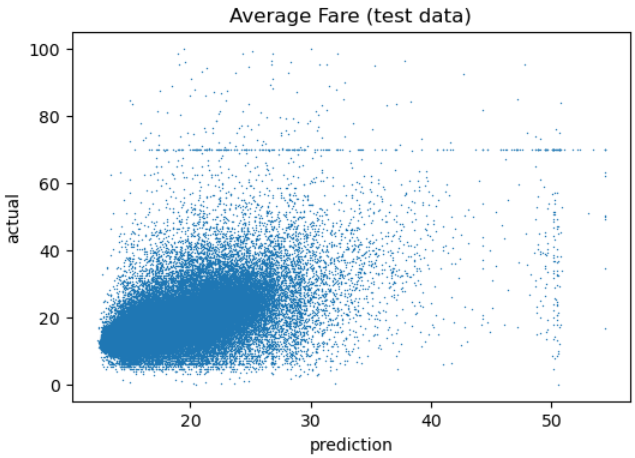
Model	Typical Error (USD)
Linear Regression	5.8 ± 0.9
Gradient Boosting	5.7 ± 0.8
Neural Network	5.9 ± 1.1

Table 3: Average prediction error for fare per trip (Mean RMSE and STD from cross validation)

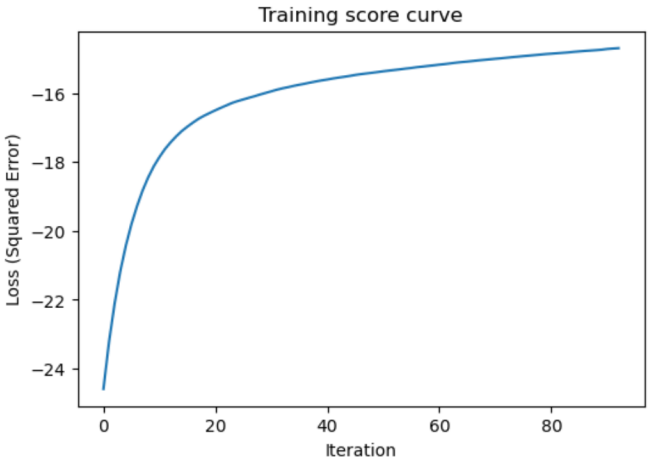
As shown in Table 3, all models achieve similar levels of accuracy, with predictions typically within around $6 per trip of the true fare. Gradient Boosting performs marginally better than the alternatives, with a slightly lower typical error and more consistent performance across different samples. While the improvement over Linear Regression and the Neural Network is small, Gradient Boosting offers a good balance of accuracy and stability, while also being faster and simpler to train.

For these reasons, and because it is significantly less costly to train than a Neural Network, Gradient Boosting is selected as the final model.

5.4 Final Average Fare Model



(a) Average Fare Model Actual vs Predicted Plot



(b) Training Score Plot

Figure 2

Dataset	Typical Error (USD)	Avg. Error (USD)
Train	5.4	3.1
Test	5.7	3.2

Table 4: Prediction accuracy of the final model (average fare per trip). Typical error refers to RMSE and average error refers to MAE

The model shows stable performance and maintains similar accuracy on new data (Table 4), indicating it can be relied on in practice.

On unseen data, predictions are within around $3.2 per trip of the true average fare (on average). This provides a useful estimate of earning potential across different locations and times. Errors are very similar between training and test data (around $5.4 vs $5.7 typical error), suggesting strong generalisation. There are limitations, as seen in figure 1a, with the model incorrectly predicting many of the fares of $70. These are mostly airport trips, which the model is not accounting for consistently.

Overall, while fare predictions are naturally more variable than demand, the model remains stable and provides a reliable basis for comparing expected earnings across zones and time periods - with the slight exception of airport trips, which are not being predicted correctly.

6 Feature Importance

Both of the models evaluated in the previous section use a large number of data features to make predictions. From these models, feature importance readings can also be generated, allowing inspection of which features were most impactful to the models. Table 5 below shows the top 5 most important features to the demand and average fare predictive models respectively.

Rank	Feature
1	Trip count (same hour yesterday)
2	Total trip duration (same hour previous day)
3	Total trip distance (same hour previous day)
4	Trip count (1 hour ahead, yesterday)
5	Total trip duration (same hour yesterday)

Rank	Feature
1	Pickup zone (for specific zones)
2	Hour of day
3	Average fare (same hour yesterday)
4	Average fare (1 hour ahead, yesterday)
5	Average fare (previous day average)

Table 5: Demand (L) and Average Fare (R) Top 5 Feature Importances

What drives demand? In terms of the model, we can determine from table 5 that demand is predicted largely through analysis of recent demand. This makes sense as demand showed temporal patterns throughout the day and week. This is consistent with the results in figure 3. Additionally, the presence of lagged trip duration and trip distance in the most important features (which are both proxies for certain zones) implies that demand is also

impacted greatly by geographical location.

What impacts average fare? Average fare amount for a given zone-hour varies across pickup zones. Some zones have extremely high or low average fares, and the model predicts these zones strongly. Where the average fare is less extreme, the model uses pickup zone to a lesser extent. As in table 5, the time of day also matters and, to a lesser extent this time, the lagged average fare performance.

These models were built to include as much information as possible as to try and learn the small trends that exist in a dynamic, complex system such as taxi demand in NYC. As a result, they contain some features that are similar or otherwise have little impact on the model. A practical improvement that could be made to these models is the removal of a number of features that add training and maintenance costs. This would make them less costly to train and run, and it is possible that they would generalise even better to unseen data.

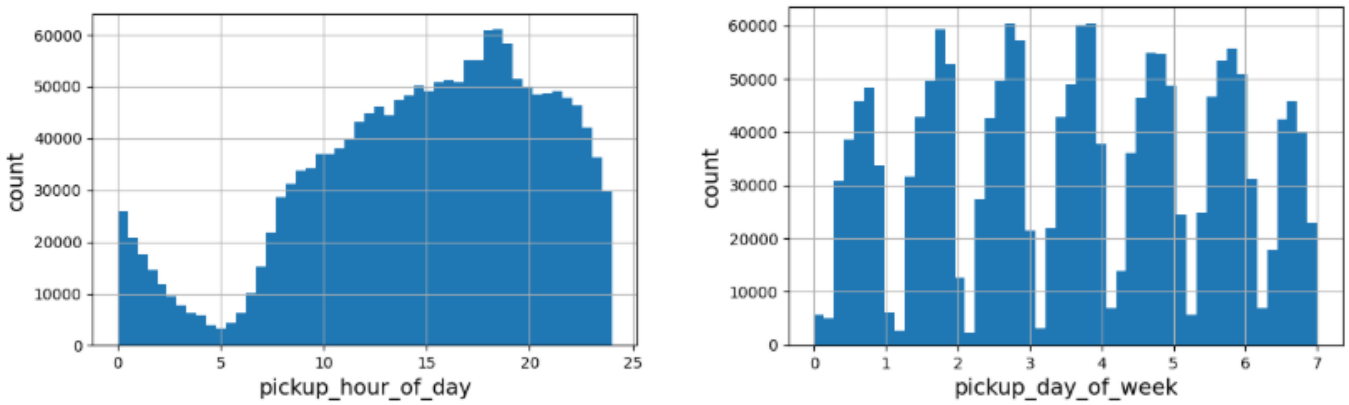
7 Data Observations and Recommendations

7.1 Demand Patterns in Time

Demand exhibits strong and consistent hourly and weekly patterns, as in figure 3:

- Peaks around 6pm (rush hour)
- Lows around 5am
- Highest activity on Wednesdays and Thursdays
- Trip duration is greatest in the afternoon (increased footfall and rush hour)

In addition, variation across the year is less significant.



(a) Trip count vs Hour of Day

(b) Trip Count vs Day of Week

Figure 3

Recommendation: Use a fixed weekly schedule as a baseline. Increase driver availability during evening peak hours and on high demand weekdays. This accounts for increased demand and longer trip durations during congested periods.

Demand persists later into the night and early morning on weekends compared to weekdays.

Recommendation: Extend driver coverage into early morning hours (2–6am) on weekends. Weekend behaviour differs from commuting patterns and is likely driven by *leisure activity*, requiring a different driver allocation.

7.2 Geographical Demand Distribution

Trip distribution is highly concentrated geographically. A small number of zones account for a large share of trips:

- Upper East Side South (237), Midtown Centre (161), and Upper East Side North (236) account for nearly 15% of demand.
- Inwood Hill Park (128), Highbridge Park (120), and Liberty Island (105) have very little demand.

Recommendation: Pre-position drivers in proportion to demand (as a baseline). Prioritise high-activity zones and reduce presence in low-demand areas to minimise passenger wait times and improve driver utilisation. One method

of doing this would be to run the demand model continuously a day in advance and then assign drivers to zones proportionally to the predicted demand in each. Then, you can deploy additional drivers if necessary, for example if there is bad weather or a big event on in the city.

7.3 Demand and Profitability

While average fare amounts do differ across zones (highest is Randalls Island (194), lowest is Upper East Side South (237)) it is instead trip count that dominates total fare revenue:

- Midtown Centre (161), Time Square (230), Upper East Side North South (237), Upper East Side North (236), Madison Square (186) have the highest total fare revenue.
- Ellis Island (105), Highbridge Park(120), Inwood Hill Park (128), Marble Hill (153), Roosevelt Island (202) have the lowest total fare revenue.

As seen, the top performing zones in terms of fare revenue are generally the same as those with the highest trip count. A possible explanation for the high trip count is that the centrality of these locations in Manhattan lends itself to shorter and more frequent trips.

Recommendation: Do not prioritise zones with greater average fare. The accumulation of trips is the greatest earner. Expected fare revenue per zone-hour can be estimated by multiplying together model outputs for predicted trip count and average fare. This provides a prediction of total (fare) earnings potential in each zone at a given time. This allows the models detailed in this report to be used to predict the most profitable zones.

7.4 Tip Distribution

There is disparity in potential tip earnings across zones, with some areas consistently generating higher tips and tip-to-fare ratios.

Zone (Code)	Average Tip Amount ($)
Randalls Island (194)	7.74
Liberty Island (105)	5.37
Battery Park City (13)	3.91
World Trade Centre (261)	3.64
Battery Park (12)	3.56

Table 6: Top 5 pickup zones by average tip amount

Recommendation: Rotate driver shifts to ensure equal exposure to high-tipping areas (see Table 6). This supports fairness, driver retention, and operational stability.

7.5 Improving Data Collection

As noted above, data is missing from rides recorded by vendors 1, 2, and 6. While this does not prevent analysis, missing fields such as passenger count or congestion surcharge reduce confidence in the results. A key recommendation is to work with vendors to ensure more complete data recording. Additionally, some useful data is not collected at all. For example, it is not possible to measure how long taxis remain inactive between trips. Since reducing idle time is important for maximising revenue, recording this information would be valuable. This would require linking trips to a unique cab ID. This would allow analysis of driver activity and help identify where revenue is lost due to inactivity.

8 Conclusion

This project has produced models that predict trip counts and average fare rates - in each zone, for a given hour - approximately a day in advance. The above analysis has demonstrated that the models do have predictive value, despite the targets being inconsistent and difficult to predict, and therefore are useful to your ride-hailing service. This report has described how, using the models, you might optimally allocate drivers to areas that are likely to see the most demand, and how you might also take expected fare earnings into account (using the 2 models together). Finally, by analysis of the dataset, we have suggested methods for maximising business effectiveness, such as optimising driver tip allocation and developing potential for future analysis by improving data collection.